

Diffusion-Augmented Deep Learning for CRC Tissue Classification

Benchmarking ResNet-50, ViT-B/16 & ConvNeXt with Stable Diffusion Augmentation

Sharath Chandra Shankar · NCT-CRC-HE Dataset · May 2026

98.9%

Peak accuracy

9 classes

Tissue types

6 runs

MLflow tracked

What we'll cover

01

The Problem

Colorectal cancer, class imbalance, data scarcity

02

The Dataset

NCT-CRC-HE: 7,180 H&E patches, 9 tissue classes

03

Diffusion Augmentation

Stable Diffusion img2img + CLIP quality filter

04

Three Architectures

CNN vs ViT vs Hybrid — what makes each different

05

Results

98.9% accuracy, ablation study, the surprising finding

06

Production System

FastAPI + Streamlit + GradCAM + Uncertainty

01

The Problem

Why automated histology classification matters

Colorectal cancer: the clinical context

#3

Most common cancer globally

#2

Cause of cancer-related mortality

10+ yrs

Training to become a pathologist

3.95x

Class imbalance ratio in our dataset

What this project solves

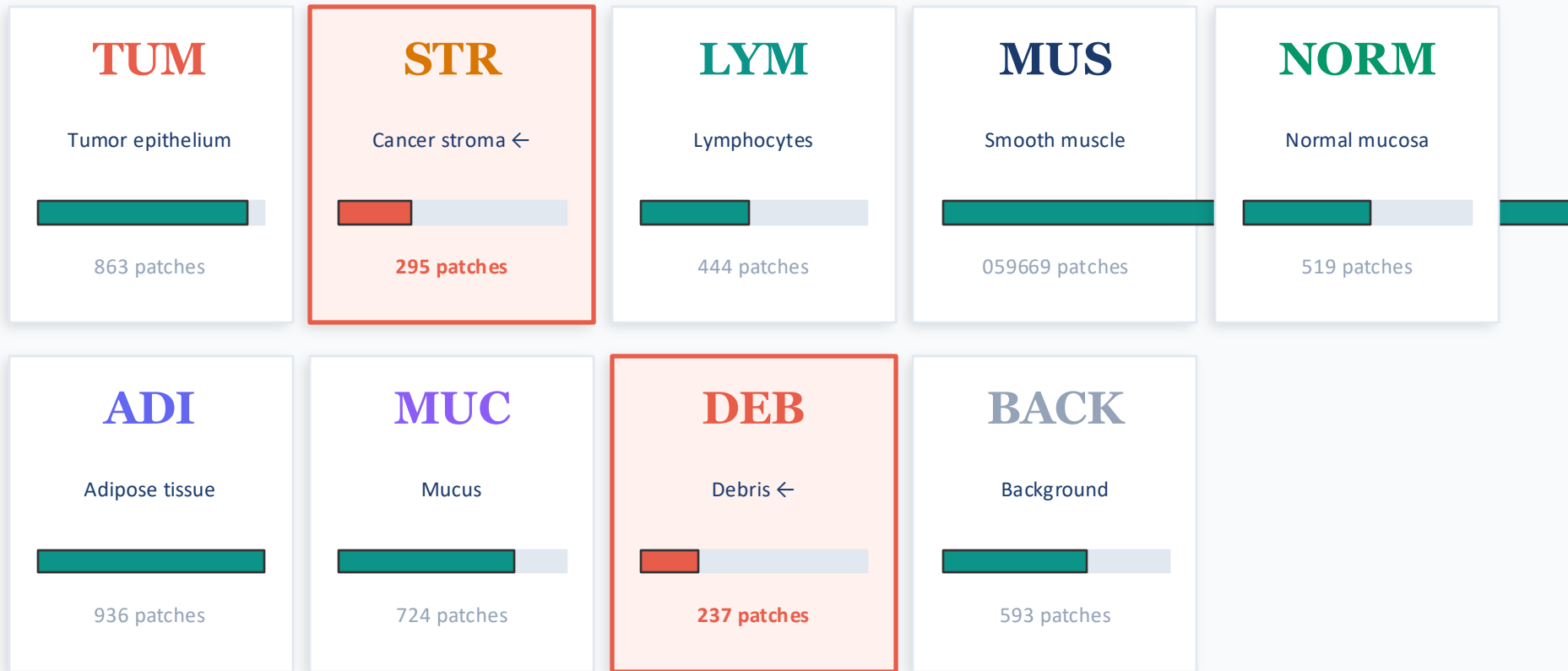
- Automate 9-class tissue classification on H&E patches
- Address class imbalance with generative augmentation
- Compare CNN vs ViT vs Hybrid architectures rigorously
- Provide explainability (GradCAM, attention maps)
- Quantify uncertainty for safe clinical deployment

02

The Dataset

NCT-CRC-HE: 7,180 H&E patches from 50 CRC patients

9 tissue classes — each tells a different biological story



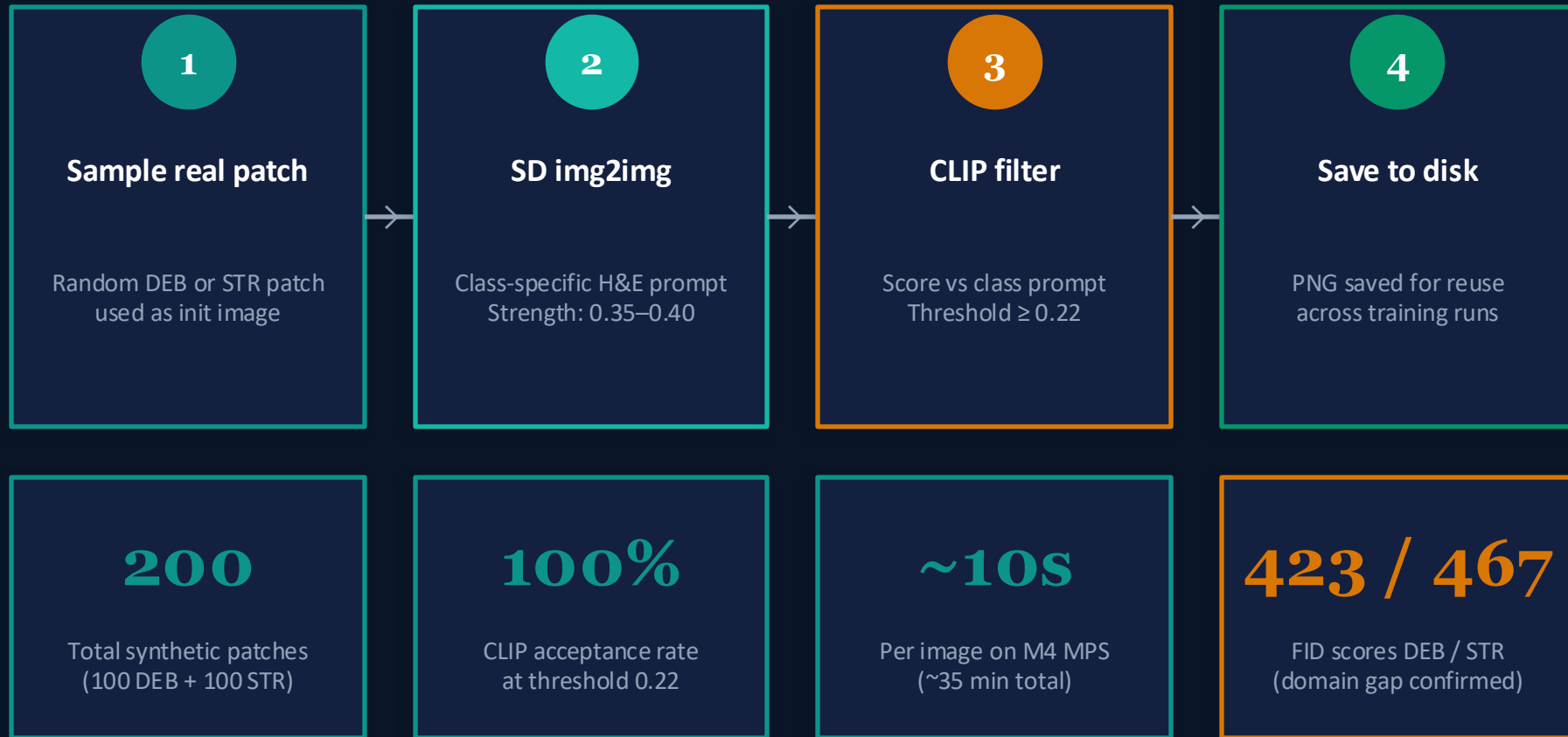
← minority classes targeted for diffusion augmentation

03

Diffusion Augmentation

Stable Diffusion img2img + CLIP quality filtering on Apple M4

The generation pipeline



04

Three Architectures

Why each model is fundamentally different

CNN vs ViT vs Hybrid — a fundamental design comparison

ResNet-50

CNN

25M params

STR: 92.1%

Strong spatial
inductive bias

*Sliding 3×3 filters
detect local patterns*

- Locality assumption
- Translation invariant
- Data efficient
- Fast stage 1 convergence

ViT-B/16

Transformer

86M params

STR: 95.2%

No spatial
inductive bias

*196 patches attend
to each other globally*

- Like LLMs for images
- Global context natively
- Best STR accuracy
- Needs more epochs

ConvNeXt

Hybrid

89M params

STR: 92.1%

CNN structure +
transformer design

*Depthwise conv +
LayerNorm + GELU*

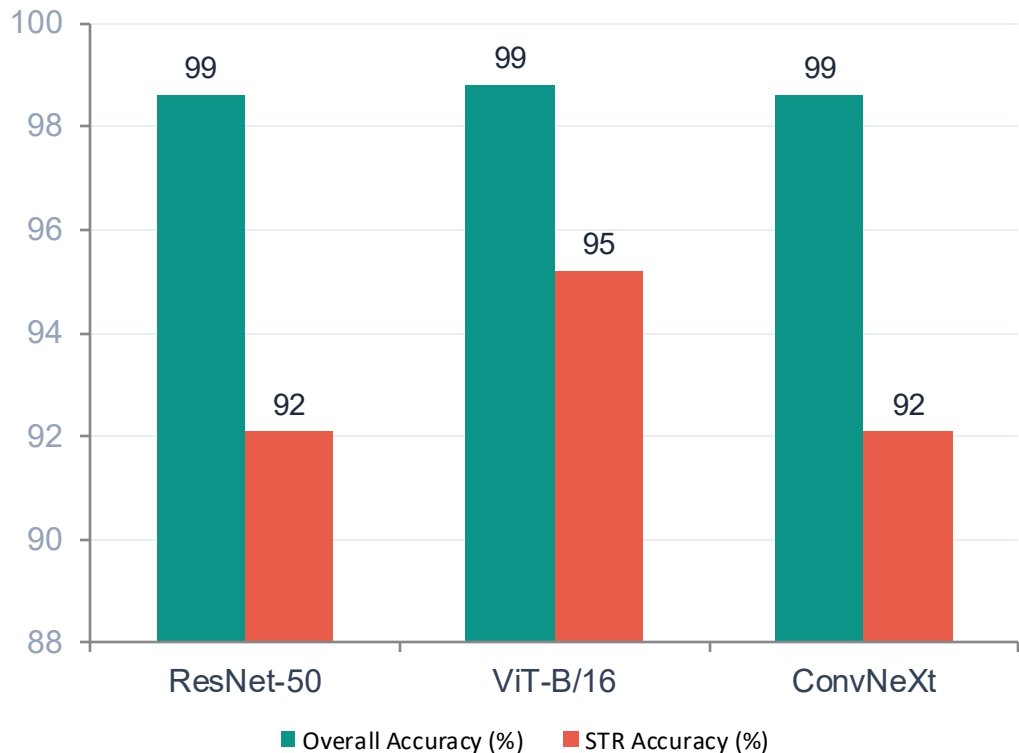
- Best of both worlds
- LayerNorm sensitive
- Hurt by synthetic data
- Matches ResNet baseline

05

Results

98.9% peak accuracy — and a surprising ablation finding

Baseline benchmark — all three models, test set performance



ViT wins on STR

95.2% vs 92.1% for CNN & ConvNeXt. Global attention captures distributed fibrous patterns better than local filters.

DEB identical across all

98.0% for all three models. Debris is a local texture problem — no architectural advantage.

Similar overall accuracy

All models within 0.2% overall. The real story is in the per-class breakdown, not the headline number.

Ablation: what synthetic augmentation actually did

ResNet-50

STR accuracy

92.1% ↑ 95.2%

baseline

+synthetic

+3.1% HELPED

ViT-B/16

STR accuracy

95.2% → 95.2%

baseline

+synthetic

0.0% NO EFFECT

ConvNeXt

STR accuracy

92.1% ↓ 90.5%

baseline

+synthetic

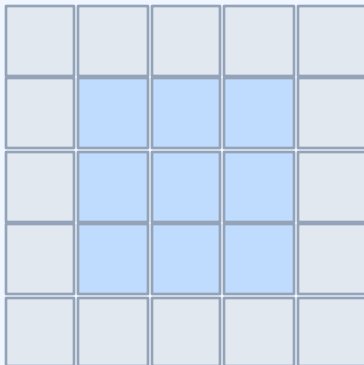
-1.6% HURT

Key finding: diffusion augmentation is architecture-dependent. It helps CNN (closes the gap with ViT), makes no difference for ViT (already solved by global attention), and hurts ConvNeXt (distribution-sensitive hybrid design).

Why ViT outperforms CNN on cancer-associated stroma

ResNet-50 (CNN)

Sees a 3×3 pixel window at a time



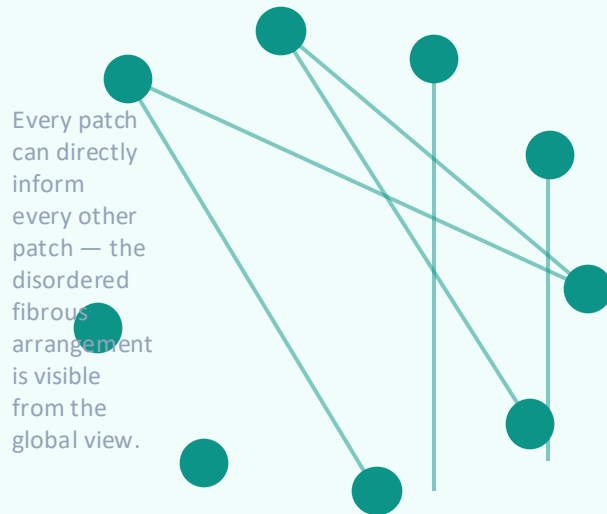
3x3 receptive field

STR diagnosis requires seeing the disorganised fibrous arrangement across the **WHOLE** patch — not just a 3×3 local window.

CNN result: 92.1% STR

ViT-B/16 (Transformer)

All 196 patches attend to each other



Every patch can directly inform every other patch — the disordered fibrous arrangement is visible from the global view.

ViT result: 95.2% STR (+3.1%)

06

Production System

FastAPI + Streamlit + Docker + GradCAM + Uncertainty

What the live demo shows — 4 outputs from one image upload

Predicted class

TUM — Colorectal adenocarcinoma
94% confidence · 35ms inference

GradCAM heatmap

Red regions = where model focused
Gradient-weighted class activation map

MC Dropout uncertainty

20 stochastic passes · std = 0.03
✓ Low uncertainty — confident prediction

ViT attention map

12-head entropy analysis
Head 01: 4.14 (diffuse) → global context

Key engineering challenges solved

01 10GB dataset download

Fix: `Explicit split='crc_val_he_7k'` — HF repo bundles 3 splits silently

02 MPS batch crash in SD

Fix: One image at a time through SD pipeline — MPS doesn't support batch attention

03 GradCAM tensor error

Fix: `.cpu().detach()` before `.numpy()` — MPS tensors cannot convert directly

04 ViT attention all 1.0

Fix: Monkey-patched `MultiheadAttention.forward` to force `need_weights=True`

05 models namespace conflict

Fix: `sys.path.insert(0, project_root)` — conda had a system-level models package

06 MLflow path spaces

Fix: Quoted the URI: `mlflow ui --backend-store-uri "file://$(pwd)/..."`

Conclusions — what we found and what it means

1 Architecture > augmentation for global classes

ViT's self-attention solves STR without any synthetic data. Architectural choice is a stronger lever than data augmentation for classes defined by global tissue arrangement.

2 Diffusion augmentation is architecture-dependent

+3.1% for CNN, 0% for ViT, -1.6% for ConvNeXt. Generative augmentation must be validated per-architecture before deployment.

3 Uncertainty quantification enables safe deployment

MC Dropout flag_for_review creates a human-in-the-loop system. A model that knows when it doesn't know is clinically safer than one that's always confident.

4 Production-ready end to end

FastAPI + Streamlit + MLflow + Docker. 6 trained models, GradCAM + attention maps, full benchmarking suite.

Thank you

The system is live at localhost:8501

98.9%

Peak test accuracy

+3.1%

STR gain (CNN +synth)

6 runs

MLflow tracked

200

Synthetic patches

4 APIs

FastAPI endpoints

Dataset: owkin/nct-crc-he · Stack: PyTorch · diffusers · MLflow · FastAPI · Streamlit · Docker

Questions?